

TRAVEL, TOURISM &
HOSPITALITY - CUSTOMER
RETENTION AND DYNAMIC
PRICING ANALYSIS
FINAL PROJECT REPORT

1. Problem Statement :

The travel and hospitality industry faces major challenges due to fluctuating demand, unpredictable customer behavior, and high cancellation rates. Hotels and travel agencies often lose significant revenue because traditional booking systems cannot accurately predict cancellations or dynamically adjust prices according to seasonal demand.

2. Tools Used :

- Power BI - Dashboard creation & visualization
- Excel - preprocessing
- Visual Studio – For Data cleaning

3. Dataset Description :

The dataset contains hotel booking information including:

- Hotel
- lead_time
- arrival_date_month
- stays_in_weekend_nights
- stays_in_week_nights
- adults
- childrens
- country
- market_segment
- distribution_channel
- adr
- reservation_status

Dataset Size:

- Approximately 119,000 records
- Multiple categorical and numerical variables

4. Data Cleaning Steps :

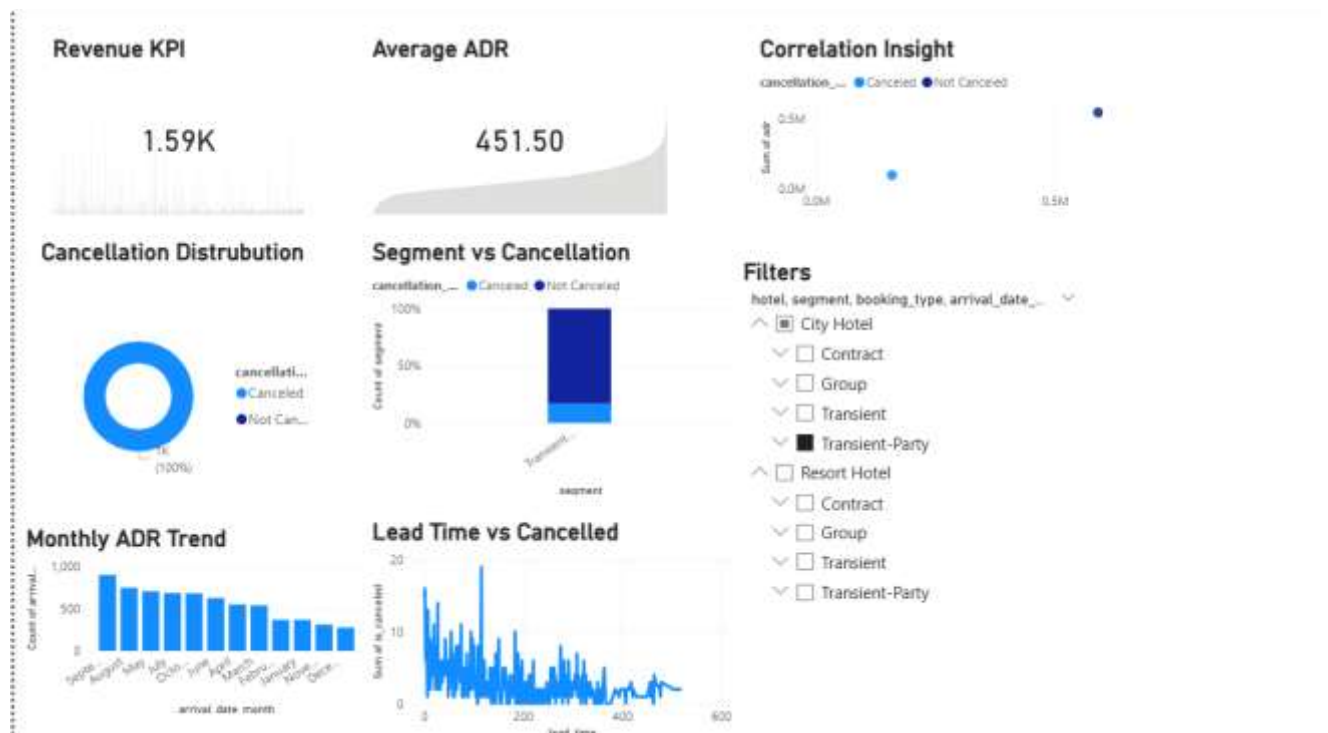
The data cleaning process is the most critical stage of this project because hotel booking datasets usually contain:

- Missing values
- Incorrect Data Type
- Duplicate Record
- Extreme Outliner
- Inconsistent categories
- Invalid booking values

A clean dataset ensures:

- Accurate EDA
- Reliable Cancellation Prediction
- Better Customer Segmentation
- Correct Pricing Insight

5. Dashboard Screenshot



6. Future Enhancement :

- Advanced ML models (XGBoost, Random Forest)
- Real-time pricing engine
- AI-based recommendation systems
- Cloud deployment
- Automated retention campaigns

7. Business Insight :

The Travel, Tourism, and Hospitality Customer Retention and Dynamic Pricing Analysis project provides valuable business insights into customer booking behavior, cancellation trends, and seasonal demand patterns. The analysis shows that factors such as long lead times, flexible deposit policies, high room prices, and specific customer segments significantly influence booking cancellations. Seasonal trends reveal that hotel demand and pricing fluctuate throughout the year, enabling hotels to apply dynamic pricing strategies during peak and off-peak periods. Customer segmentation helps identify high-value groups such as corporate and repeat travelers, allowing targeted marketing and retention campaigns. Additionally, predictive modeling helps hotels identify high-risk cancellations in advance, enabling proactive actions to reduce revenue loss. Overall, the project supports data-driven decision-making to improve occupancy, optimize pricing, enhance customer retention, and increase overall hotel profitability.

8. Recommendation :

Hotels should implement dynamic pricing strategies based on seasonal demand, occupancy, and customer behavior to maximize revenue and improve room utilization. The business should reduce cancellation rates by introducing deposit policies, personalized retention offers, and reminder notifications for high-risk bookings. Hotels should also focus on promoting direct bookings through loyalty programs and exclusive discounts to reduce dependency on third-party platforms. Additionally, targeting valuable customer segments such as corporate travelers and repeat guests with customized marketing campaigns can improve customer retention, occupancy, and overall profitability.

9. Conclusion :

This project successfully demonstrates how data analytics and machine learning can transform decision-making in the hospitality industry.

Through extensive exploratory data analysis, customer segmentation, and predictive modeling, the project identifies key cancellation drivers and seasonal pricing patterns.

The developed dashboards and predictive insights provide hotel management teams with actionable intelligence for improving customer retention, maximizing revenue, and enabling future AI-driven pricing systems.

The project establishes a strong analytical foundation for advanced hospitality revenue management solutions.